

Problem

In a system, the input signal $x(t)$ is amplitude modulated by $m(t) = 2 + \cos \omega_0 t$. The response $y(t) = m(t)x(t)$. Find $Y(\omega)$ in terms of $X(\omega)$.

Step-by-step solution

Step 1 of 2

Consider a system, in which input signal $x(t)$ is amplitude modulated by the signal $m(t) = 2 + \cos \omega_0 t$ and its response is $y(t) = x(t)m(t)$.

Calculate the Fourier transform for the response $y(t)$.

$$\begin{aligned} Y(\omega) &= \mathcal{F}[x(t)m(t)] \\ &= \mathcal{F}[x(t)(2 + \cos \omega_0 t)] \\ &= \mathcal{F}\left[x(t)\left(2 + \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right)\right] \\ &= 2\mathcal{F}[x(t)] + \frac{1}{2}\mathcal{F}[x(t)e^{j\omega_0 t}] + \frac{1}{2}\mathcal{F}[x(t)e^{-j\omega_0 t}] \end{aligned}$$

Step 2 of 2

Write the Fourier transform pairs that are used to find the $Y(\omega)$.

By the definition of Fourier transform,

$$\mathcal{F}[x(t)] = X(\omega)$$

By frequency shift property,

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

Apply the Fourier transform pairs to determine the Fourier transform for $y(t)$.

$$\begin{aligned} Y(\omega) &= 2(X(\omega)) + \frac{1}{2}F(\omega - \omega_0) + \frac{1}{2}F(\omega + \omega_0) \\ &= 2(X(\omega)) + 0.5F(\omega - \omega_0) + 0.5F(\omega + \omega_0) \end{aligned}$$

Thus, the $Y(\omega)$ in terms of $X(\omega)$ is $\boxed{2(X(\omega)) + 0.5F(\omega - \omega_0) + 0.5F(\omega + \omega_0)}$.